9A Lesson 3

Probability and statistics

Year 9 Science · 9A Genetics and evolution · Student Book pages 10 to 11

Name _____ Class _____ Date _____

Today I can:

- ☐ I can explain what probability is and write a probability as a percentage, a fraction and a decimal.
- ☐ I can calculate an experimental probability from results and compare it with a theoretical probability.

Key words

statistics collecting, analysing and presenting data
sample a small part of a group that is measured to estimate the whole group
mean the sum of the values divided by the number of values
median the middle value when the values are in order
mode the most common value in a set
probability the chance of something happening
percentage a number out of 100, shown with the % sign
fraction a part of a whole, written as one number over another, e.g. 1/4
decimal a number written with a decimal point, e.g. 0.25
experimental probability a probability worked out from the results of an experiment
theoretical probability a probability worked out by reasoning about all the possible outcomes

● Core Everyone does this section.

1. Match each scientific term to its definition. Draw a line.

mean	the most common value in a set
median	the chance of something happening
mode	the sum of the values divided by the number of values
probability	the middle value in a set of values in order

2. Complete the probability scale. Fill in the missing values.

Percentage	Fraction	Decimal	Word
0%	0	0	impossible
25%	1/4		
50%		0.5	even chance
75%			likely
100%	1		certain

3. Two dice investigation. Throw two dice, add the numbers, and tally the score. Do 50 throws.

a) Complete the table. Experimental probability = frequency divided by 50.

Score	Tally	Frequency	Fraction (out of 50)	Decimal	Percentage
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					

b) Which score was the mode (the most common) in your results?

●● Stretch When Core is finished.

S1. A weather forecast says there is a 5% chance of rain in your area between 11:00 and 12:00. Tick the box that gives the correct meaning.

- ☐ A Rain is predicted to fall for 5% of the time between 11:00 and 12:00.
- ☐ B If that hour happened 100 times, it would rain on about five of them.
- ☐ C Rain is predicted to fall at five different times between 11:00 and 12:00.
- ☐ D 5% of the rain predicted for tomorrow will fall between 11:00 and 12:00.

S2. A bag contains 50 beads: 20 red, 10 green, 5 blue and 15 yellow. Bilal takes one bead without looking.

a) Give the probability that the bead is blue as a fraction, a decimal and a percentage.

b) Calculate the probability that the bead is not red. Give your answer as a percentage.

S3. The theoretical probability of a score of 7 with two dice is 6/36, which is about 17%. Compare this with your experimental probability for 7. Suggest a reason for any difference.

●●● Challenge Explain in full sentences.

C1. Achondroplasia is an inherited condition. If neither parent has it, the theoretical probability of a child having it is 0.003%. If one parent has it, the probability is 50%. If both parents have it, the probability is 75%.

a) Write the probability for a child with one affected parent as a decimal and as a fraction.

b) A couple (one parent affected) has two children without the condition. Explain whether their third child is more likely to have it.

c) Explain why these are theoretical probabilities and not experimental probabilities.

C2. Calculate the theoretical probability of throwing a number less than three with one dice. Give your answer as a fraction, a decimal and a percentage. Then explain why an experimental probability from 100 throws is more reliable than one from 10 throws.

My thinking (any language)

A friend says: 'I rolled a six three times in a row, so the next roll will definitely not be a six.' Do you agree? Explain your thinking.

Word bank statistics · sample · mean · median · mode · probability · percentage · fraction · decimal · experimental probability · theoretical probability

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Exit ticket 9A Lesson 3: Probability and statistics Name _____

1. What is probability?

2. Write 5% as a decimal.

3. A bag has 20 red, 10 green, 5 blue and 15 yellow beads. Give the probability of taking a blue bead as a fraction and as a percentage.

Today I can: Goal 1 ☐ Goal 2 ☐